Proposal for an alternative approach for estimating CO_2 from fossil fuel use.
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CO_2 emissions from fossil fuel use in the Soil Enrichment Protocol (Revised)

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1. Description of Proposal

The Soil Enrichment Protocol includes change in fossil fuel usage as a land management practice eligible for consideration in soil enrichment projects (SEP v1.1, section 2.2.1, Appendix B). The SEP uses Equations 5.28 and 5.29 to calculate average non-reversible CO_2 emissions reductions from fossil fuel use (Section 5.4.3). These calculations are based on consumption of fossil fuel stratified by vehicle/equipment type and quantified using the emission factor of the type of fossil fuel combusted.

In Indigo's experience, farmers are limited in their ability to provide data on the specific pieces of equipment in use on their farm, much less the duration that each piece of equipment is used throughout the cultivation cycle and/or the amount of fuel consumed on each field by each type of equipment during the cultivation cycle. Previously, Indigo has not accounted for fossil fuel emissions, but rather considered them *de minimis,* largely due to these constraints in data availability and the fact that our practice changes overall result in emission reductions (so exclusion is conservative). To overcome these challenges for accounting CO_2 emissions (and reductions) due to fossil fuel during farming operations, Indigo is proposing an alternative approach which is a minor variation of the current SEP equations. Instead of focusing on types of vehicles and equipment, the equation is adjusted to focus on management activities and the estimated fuel use (and changes in fuel use) related to land management events. This allows fuel emissions to be estimated using the management data that are already being collected for the SEP.

In this approach, the average non-reversible CO_2 emissions reductions from fossil fuel are estimated by subtracting CO_2 emissions from fossil fuel use in the project scenario from that of the baseline for the corresponding agronomic practices. [The term "agronomic practices" includes individual field operations such as plowing, planting, fertilizing, etc.,]. Thus, fossil fuel use for agronomic practices in the baseline or the project scenarios can be estimated using average fuel use for individual field operations adopted during a cultivation cycle. We have provided a recommended source for values for the average fuel used for individual field operations. For simplicity, it is assumed that the relevant equipment is using diesel fuel, as this is by far the most common in agricultural operations. If individual farms have more efficient alternatives (e.g., electric vehicles) they still have the option of the vehicle-specific fossil fuel equations.

Indigo is seeking the Reserve's approval for using this alternative approach to estimate non-reversible CO_2 emissions reductions from fossil fuel in the third verification of Project CAR1459, as well as its inclusion as an additional approach in the SEP that is usable by all projects. This can be achieved by editing Eq. 5.29 to estimate CO2 emissions from fossil fuel based on agronomic practice. This does not necessitate any change to Eq. 5.28.

2. Proposed equations

The proposed equation (5.29b) to calculate non-reversible CO_2 emission reductions from fossil fuel consumption based on agronomic practices is shown below (Figure 1).

Equation 5.29b Carbon Dioxide emissions from fossil fuels based on agronomic practice

$$\overline{CO2_NR_{s,t}} = \sum_{e}\left(\overline{FFC_{e,s,t}} \times EF_{co_2}\right)$$

Where,

			Units
$\overline{CO2_NR_{s,t}}$	=	Average carbon dioxide emissions from diesel fuel use per unit area in stratum s in cultivation cycle t	tCO_{2e}/acre
$\overline{FFC_{e,s,t}}$	=	Average consumption of diesel fuel use per unit area stratum s in an agronomic practice e in cultivation cycle *t*	gal/acre
EF_{CO_2}	=	Emission factor for diesel fuel	tCO_{2e}/gal

Figure 1: Proposed equation to estimate CO_2 emission from fossil fuel during agronomic practices. Changes to the original equation 5.29 are highlighted in red.

As indicated, there are no changes to the original Equation 5.28 of the SEP, and hence the output can be directly used in Equation 5.6 to calculate the total non-reversible emission reductions for a reporting period. The parameters necessary to calculate baseline and project emissions are provided in Table 1.

Table 1: Proposed parameters from equation 5.29b

Equation	**Parameter**	**Proposed Definition**	**Data Unit**	**Calculated (c) Measured (m) Reference (r) Operating records (o)**	**References**
5.29b	$\overline{CO2_NR_{s,t}}$	Average carbon dioxide emissions from diesel fuel use per unit area for stratum s during cultivation cycle t	$tCO_{2e}/acre$	c	
5.29b	$\overline{FFC_{e,s,t}}$	Average consumption of diesel fuel per unit area for stratum s in agronomic practice e in cultivation cycle t	$gal/acre$	r	Estimates for fuel use under various field operations can be found in the attached standardized look up table.
5.29b	EF_{CO2}	Emission factor for diesel fuel	tCO_{2e}/gal	r	Greenhouse Gases Equivalencies Calculator - Calculations and References | US EPA

We are submitting a standardized table of default values for estimated fossil fuel (diesel fuel) consumption for various field operations (Table 2). We chose to use only diesel fuel data as almost all the farm equipment used for field operations operate on diesel fuel. The standardized look-up table has been adapted from a Colorado State University Extension Fact Sheet that includes a comprehensive list of implements and their fuel use (Downs and Hansen. 1998). Indigo recommends that in using this standardized look-up table, Project Developers must:

1. Provide a justification for how specific implements, events, or methods were mapped to operations in look up

table. Justifications may come from literature or a subject matter expert and

2. If data are not collected on non-fertilizer chemical application events, include one additional sprayer operation in the project condition when tillage is reduced between the baseline and project conditions.

3. If a tillage pass was not used to terminate a cover crop and information is not collected on non-fertilizer chemical applications, include one additional sprayer operation on the date of cover crop termination.

Table 2: Diesel use for farm operations*

Operation**	**Gallons of diesel per acre**
Shred stalks	0.72
Plow 8 inches deep	1.68
Heavy offset disk	0.95
Chisel plow	1.10
Tandem disk, stalks	0.45
Tandem disk, chiseled	0.55
Tandem disk, plowed	0.65
Field cultivate	0.60
Spring-tooth harrow	0.40
Spike-tooth harrow	0.30
Mulch treader	0.30
Rod weeder	0.30
Sweep plow	0.60
Cultivate row crops	0.45
Rolling cultivator	0.35
Rotary hoe	0.25
Anhydrous applicator	0.65
Planting row crops	0.50
No-till planter	0.35
Till plant (with sweep)	0.40
Grain drill	0.35
Combine, small grains	1.00
Combine, beans	1.10
Combine, corn and grain sorghum	1.60
Corn picker	1.15
Mower (cutterbar)	0.35
Mower conditioner	0.60
Swather	0.55
Rake, single	0.25

Table 2: Diesel use for farm operations*

Operation**	**Gallons of diesel per acre**
Rake, tandem	0.15
Baler	0.45
Stack wagon	0.50
Sprayer***	0.10
Rotary mower	0.80
Haul small grains	0.60
Grain drying	6.0
Forage harvester, green forage	0.95
Forage harvester, haylage	1.25
Forage harvester, corn silage	3.60
Forage blower, green forage	0.35
Forage blower, haylage	0.25
Forage blower, corn silage	1.40
Forage blower, high-moisture ear corn	0.45

*Note: Adapted from Downs, H., and R. Hansen. 1998. Estimating Farm Fuel Requirements. Colorado State University Extension, Fact Sheet 5.006.
**Project Developers must provide a justification for how specific implements, events, or methods were mapped to operations in look-up table. Justifications may come from literature or a Subject Matter Expert.
***If information is not collected on non-fertilizer chemical application, include one additional sprayer operation in the project condition when tillage is reduced between the baseline and project conditions. If a tillage pass was not used to terminate a cover crop and information is not collected on non-fertilizer chemical applications, include one additional sprayer operation on the date of cover crop termination.

Summary of Changes

Section	**Date**	**Summary of Changes**
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